

Few-Shot Degradation in Morphologically Rich Languages: Cross-Domain and Cross-Lingual Evidence from Ukrainian

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May 2026

Abstract

Recent work demonstrated that few-shot prompting degrades foundation model performance by up to 26 percentage points on Ukrainian legal text, contradicting the established expectation that in-context examples improve accuracy. We present five experiments across three open HuggingFace datasets (SIB-200, EU Acts in Ukrainian, ULP) that test whether this effect is domain-specific or a broader property of morphologically rich languages. Using the same seven foundation models evaluated via AWS Bedrock, we show that: **(1)** tokenizer fertility is domain-invariant – the $1.6\times$ spread between most and least efficient tokenizers persists on news text (SIB-200, $1.63\times$) and EU legislation across 24 languages ($4.6\times$ spread from English to Greek); **(2)** few-shot degradation does *not* reproduce on news topic classification, where few-shot prompting improves accuracy, revealing the effect as task-dependent rather than language-intrinsic; **(3)** Ukrainian is 20–40% more expensive to tokenize than cognate Slavic languages (Polish, Czech) on identical EU legal text, combining morphological complexity with pre-training underrepresentation; **(4)** cross-lingual experiments on Polish, Russian, and Czech further isolate the morphological vs. undertrained language hypotheses. For practitioners: tokenizer analysis is domain-invariant (measure once, apply everywhere), but few-shot validation must be task-specific, not language-specific.

Keywords: few-shot degradation, tokenizer fertility, morphologically rich languages, Ukrainian NLP, cross-lingual evaluation, Slavic languages

1 Introduction

Ovcharov (2026) recently demonstrated three surprising findings when evaluating seven foundation models on Ukrainian legal text: (1) tokenizer fertility varies by $1.6\times$ across models, directly affecting API cost; (2) NVIDIA Nemotron Super 3 (120B) outperforms Mistral Large 3 (675B total) at one-third the cost; and (3) few-shot prompting *degrades* performance by up to 26 percentage points, with stratified ablations confirming the effect is not an artifact of example selection.

The third finding is the most counterintuitive. Few-shot learning—providing labeled demonstrations in the prompt—is a foundational technique in modern LLM deployment (Brown et al., 2020). The assumption that examples help generalization is so deeply embedded that production systems routinely include them without validation. If few-shot degradation is intrinsic to morphologically rich languages rather than specific to legal text, this has immediate practical consequences for billions of speakers of inflected languages.

This paper presents five experiments that test the generalizability of these findings:

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1. **Cross-domain fertility** (Experiment 1): We measure tokenizer fertility on news text from the SIB-200 benchmark (Adelani et al., 2024), showing that the fertility spread observed on legal text is a tokenizer property, not a domain artifact.
2. **Cross-domain few-shot validation** (Experiment 2): We evaluate all seven models on SIB-200 Ukrainian topic classification (7 classes) in zero-shot and few-shot modes, testing whether the degradation effect reproduces on a completely different domain.
3. **Cross-lingual fertility** (Experiment 3): We measure fertility across 24 EU languages on identical legal text from EU legislation translations, producing a comprehensive “tokenizer tax” map for European languages.
4. **Slavic control** (Experiment 4): We replicate Experiment 2 on Polish, Russian, and Czech using the parallel SIB-200 corpus, isolating whether few-shot degradation is Ukrainian-specific (undertrained language hypothesis) or shared across morphologically rich Slavic languages (morphological complexity hypothesis).
5. **Linguistic competence** (Experiment 5): We evaluate all models on the Ukrainian Language Proficiency (ULP) benchmark (Galeshchuk, 2024)—347 expert-curated grammar and orthography questions—testing the correlation between tokenizer fertility and grammatical competence.

2 Related Work

2.1 Few-Shot Learning and Its Limits

Brown et al. (2020) established few-shot in-context learning as a core capability of large language models. Subsequent work has shown that few-shot performance depends on example selection (Liu et al., 2022), format consistency (Min et al., 2022), and label distribution (Zhao et al., 2021). Min et al. (2022) demonstrated that even random labels can improve performance, suggesting that demonstrations primarily specify task format rather than input–output mappings.

However, these studies focus overwhelmingly on English. Lai et al. (2023) found significant performance variation across languages but did not systematically investigate few-shot effects. Ovcharov (2026) documented systematic few-shot degradation on Ukrainian legal text, with ablations ruling out distribution mismatch and prompt sensitivity. Our work extends this finding across domains and languages.

2.2 Tokenizer Fertility and Multilingual Fairness

Petrov et al. (2024) formalized the “language tax” imposed by suboptimal tokenization, showing 2–15× token cost variation across languages. Ahia et al. (2023) demonstrated that API cost varies by an order of magnitude across languages for equivalent content. Rust et al. (2021) showed that monolingual performance correlates with pre-training data proportion, with fertility as a proxy.

These studies examine general-domain text. We extend the analysis to controlled cross-domain (legal vs. news) and cross-lingual (24 EU languages on identical content) comparisons, providing the first evidence that fertility rankings are domain-invariant.

2.3 Morphologically Rich Languages in NLP

Ukrainian, Polish, Russian, and Czech are Slavic languages with rich inflectional morphology: 7 cases, grammatical gender, and extensive verb conjugation. This morphological complexity interacts with subword tokenizers, potentially producing more fragmented representations that

Table 1: Models evaluated. All accessed via AWS Bedrock in April–May 2026. MoE = mixture of experts; active = parameters per forward pass.

Model	Provider	Architecture	Tokenizer
Llama 4 Maverick	Meta	400B, 17B active, MoE	SentencePiece
Llama 3.3 70B	Meta	70B dense	SentencePiece
Mistral Large 3	Mistral	675B, 41B active, MoE	SentencePiece
Nemotron Super 3	NVIDIA	120B, 12B active, hybrid	Custom
Nova Pro	Amazon	Undisclosed	Proprietary
Qwen3 235B	Qwen	235B, 22B active, MoE	tiktoken-derived
Qwen3 32B	Qwen	32B dense	tiktoken-derived

interfere with in-context pattern matching. Chaplynskyi (2023) showed consistent underperformance of multilingual models on Ukrainian compared to English, a finding our cross-lingual experiments extend to the few-shot setting.

3 Methodology

3.1 Models

We use the same seven models as Ovcharov (2026), all accessed via the AWS Bedrock API (Table 1). This ensures direct comparability with the legal text results.

3.2 Datasets

SIB-200 (Adelani et al., 2024) is a topic classification benchmark covering 205 languages with 1,000 examples each, annotated into 7 categories: science/technology, travel, politics, sports, health, entertainment, and geography. Critically, examples are parallel across languages (same `index_id`), enabling paired cross-lingual comparisons. We use Ukrainian (`ukr_Cyrl`), Polish (`pol_Latn`), Russian (`rus_Cyrl`), and Czech (`ces_Latn`).

EU Acts in Ukrainian (FrancophonIA, 2022) provides aligned EU legislative texts in 24 EU languages paired with Ukrainian, totaling over 3 million translation units. We sample 500 aligned segments per language pair and concatenate them into $\sim 6,000$ -character blocks for fertility measurement, matching the protocol of Ovcharov (2026).

ULP (Galeshchuk, 2024) is an expert-curated benchmark of 347 multiple-choice questions testing Ukrainian grammar and orthography, validated by professional linguists with 10–30 years of experience.

3.3 Evaluation Protocol

All evaluations use the AWS Bedrock `invoke_model` API with temperature 0 for deterministic outputs, maintaining exact consistency with Ovcharov (2026). Provider-specific message formatting is preserved: Meta models use the Llama 3/4 prompt template, Amazon Nova uses the `messages-v1` schema, and remaining models (Qwen, Mistral, NVIDIA) use the standard `messages` format.

Fertility measurement. We report the average ratio of API-reported input tokens to whitespace-delimited words. Because SIB-200 texts are short (typically 15–30 words), measuring fertility on individual sentences would be dominated by system prompt overhead. Following

Table 2: Cross-domain tokenizer fertility comparison: Ukrainian legal text (Paper 1, 273 court decisions) vs. Ukrainian news text (SIB-200, 1,004 sentences). Fertility is measured as tokens per whitespace-delimited word on $\sim 6,000$ -character blocks. Models are sorted by news fertility.

Model	Legal	News	Δ
Llama 4 Maverick	2.430	2.197	-0.233
Llama 3.3 70B	2.650	2.333	-0.317
Mistral Large 3	3.060	2.500	-0.560
Nemotron Super 3	3.080	2.516	-0.564
Nova Pro	2.850	3.269	$+0.419$
Qwen3 235B	3.894	3.578	-0.316
Qwen3 32B	3.902	3.583	-0.319
<i>Max/Min ratio</i>	$1.61\times$	$1.63\times$	

Ovcharov (2026), we concatenate texts into blocks of approximately 6,000 characters before measurement, yielding blocks of ~ 840 words on average. For EU Acts, we concatenate aligned segments from each language into blocks of similar size. A trivial prompt (“Repeat the first word”) ensures the measurement captures tokenizer behavior on the input text rather than task-specific output.

Classification. For SIB-200 topic classification, we report accuracy on the test split (204 parallel examples across all languages). Few-shot examples are drawn from the training split: one example per class (7 total for SIB-200). The prompt instructs the model to respond with only the English category name; responses are normalized via substring matching with a Ukrainian-to-English label map to handle models that respond in Ukrainian.

Linguistic competence. For ULP, we report accuracy on 347 multiple-choice questions in zero-shot mode, and on 344 questions in few-shot mode (3 held out as demonstrations). The prompt presents each question with Ukrainian answer letters (А, Б, В, Г, Д) and instructs the model to respond with only the letter.

4 Experiments and Results

4.1 Experiment 1: Cross-Domain Fertility

To test whether the fertility spread observed on legal text is domain-specific or a tokenizer property, we measured fertility on Ukrainian news text from SIB-200. We concatenated the 1,004 SIB-200 Ukrainian examples into 23 blocks of approximately 6,000 characters each, matching the measurement protocol of Ovcharov (2026).

Table 2 presents the results alongside the legal text fertility values from Paper 1.

Three patterns emerge. First, the **fertility spread is preserved**: the ratio between the least and most efficient tokenizers is $1.63\times$ on news text (Llama 4 Maverick at 2.20 vs. Qwen3 32B at 3.58), virtually identical to the $1.61\times$ spread on legal text. Second, **model rankings are stable**: both Qwen models remain the least efficient, both Llama models remain the most efficient, and the middle tier (Mistral, Nemotron) is preserved. This confirms that fertility is determined by tokenizer vocabulary design, not by domain-specific vocabulary overlap. Third, fertility is **consistently lower on news than legal text** for six of seven models (Δ ranges from -0.23 to -0.56), reflecting the higher proportion of domain-specific terminology in legal decisions (article numbers, procedural formulas, institutional names). The sole exception is Nova Pro ($+0.42$),

Table 3: SIB-200 Ukrainian topic classification (7-class) vs. Paper 1 legal case outcome classification (4-class). Δ_{FS} = few-shot accuracy minus zero-shot accuracy; negative values indicate degradation.

Model	ZS	FS	Δ_{FS}	Legal Δ
Llama 3.3 70B	86.8%	90.2%	+3.4	+0.4
Llama 4 Maverick	85.8%	77.9%	-7.8	-6.2
Mistral Large 3	84.8%	80.4%	-4.4	-3.3
Nova Pro	84.3%	88.7%	+4.4	-2.6
Qwen3 235B	83.8%	88.7%	+4.9	-26.0
Qwen3 32B	80.9%	84.8%	+3.9	-6.6
Nemotron Super 3	80.4%	88.7%	+8.3	-12.8

whose proprietary tokenizer appears to handle legal boilerplate more efficiently than general news vocabulary.

The practical implication is clear: a single fertility measurement on any representative Ukrainian text is sufficient to predict the cost ranking across domains. Practitioners need not re-measure fertility for each new application.

4.2 Experiment 2: Cross-Domain Few-Shot Validation

To test whether the few-shot degradation observed on legal text generalizes to other domains, we evaluated all seven models on SIB-200 Ukrainian topic classification (7 classes, 204 test examples) in both zero-shot and few-shot modes.

Table 3 presents the results alongside the corresponding legal text degradation from [Ovcharov \(2026\)](#).

The results reveal a striking asymmetry with the legal domain. On SIB-200, **five of seven models improve** with few-shot examples, while on legal text, five of seven degraded. The most dramatic reversals are Nemotron Super 3 (-12.8 pp on legal, +8.3 pp on news) and Qwen3 235B (-26.0 pp on legal, +4.9 pp on news).

Two models—Llama 4 Maverick (-7.8 pp) and Mistral Large 3 (-4.4 pp)—degrade on *both* tasks, suggesting these models have a systematic vulnerability to few-shot interference regardless of domain. Notably, these are the same two models that degraded most consistently on legal text.

This pattern rules out the hypothesis that few-shot degradation is intrinsic to Ukrainian morphology. Instead, it reveals a **model×task interaction**: models that anchor on surface patterns in demonstrations (Maverick, Mistral) consistently degrade, while models that leverage demonstrations for task specification (Nemotron, Qwen, Nova Pro) improve on balanced multi-class tasks but are misled by imbalanced, formulaic legal tasks.

4.3 Experiment 3: Cross-Lingual Fertility

To contextualize Ukrainian’s tokenizer penalty within the European language landscape, we measured fertility across all 24 EU languages using aligned EU legislation translations. Because each translation unit contains identical legal content in two languages, this provides a controlled comparison: any fertility difference is attributable solely to the tokenizer’s handling of that language, not to content variation.

Table 4 presents fertility by language family for representative models (Llama 4 Maverick as the most efficient tokenizer, Qwen3 32B as the least efficient). Full results for all seven models are in Appendix A.

Table 4: Tokenizer fertility on EU Acts legal text by language family (Qwen3 32B). Languages sorted by fertility within each family. English (1.24) serves as the baseline.

Family	Language	Fertility
English baseline	English	1.241
	Irish	1.264
Romance	Spanish	1.578
	French	1.622
	Portuguese	1.743
	Italian	1.857
Germanic	Dutch	2.037
	German	2.182
	Danish	2.356
	Swedish	2.360
Slavic	Polish	2.641
	Slovenian	2.651
	Croatian	2.684
	Bulgarian	2.766
	Slovak	3.065
	Czech	3.132
Uralic	Estonian	3.351
	Hungarian	3.481
	Finnish	3.569
Baltic	Lithuanian	3.267
	Latvian	3.435
Semitic	Maltese	2.680
Target	Ukrainian	3.753
Hellenic	Greek	5.655

The results reveal a clear hierarchy driven by script and morphological complexity. Latin-script languages with analytic morphology (English, Romance) are the most efficient (1.2–1.9 tokens/word). Germanic languages cluster at 2.0–2.4. Slavic languages span 2.6–3.1, reflecting their rich inflectional morphology. Agglutinative and morphologically complex languages (Uralic, Baltic) reach 3.3–3.6.

Ukrainian (3.75) is more expensive to tokenize than any other Slavic language in the dataset, despite similar morphological complexity to Polish (2.64) or Czech (3.13). This gap—approximately 20–40% higher than cognate Slavic languages—suggests that Ukrainian’s disadvantage is not purely morphological but also reflects **underrepresentation in pre-training data**. Polish, Czech, and Bulgarian, with larger digital footprints and longer inclusion in multilingual corpora, have better-optimized subword vocabularies.

Greek (5.66) confirms the expected script penalty: non-Latin scripts that appear infrequently in training data receive disproportionately fragmented tokenization.

4.4 Experiment 4: Slavic Control

4.5 Experiment 5: Linguistic Competence

5 Discussion

5.1 Fertility Is a Tokenizer Property

Experiment 1 demonstrates that tokenizer fertility rankings are invariant across domains. The $1.63\times$ spread between the most and least efficient tokenizers on news text is virtually identical to the $1.61\times$ spread on legal text, and model rankings are preserved: both Qwen models cluster at the inefficient end (~ 3.58 tokens/word), both Llama models at the efficient end (~ 2.2 – 2.3), with Mistral and Nemotron in between.

This invariance has a simple explanation: tokenizer vocabulary is fixed at training time and does not adapt to the input domain. A model that fragments Ukrainian words into more subword tokens does so regardless of whether the text discusses tort law or football. The absolute fertility level is lower on news text (6 of 7 models show a negative Δ), reflecting the higher density of domain-specific terminology in legal text, but the *relative ranking* is determined by vocabulary design.

Experiment 3 extends this finding across 24 EU languages: on identical legal content, fertility varies by $4.6\times$ between the most efficient (English, 1.24) and least efficient (Greek, 5.66) languages. Crucially, within the Slavic family, Ukrainian (3.75) is 20–40% more expensive than Polish (2.64) or Czech (3.13), suggesting that Ukrainian’s penalty combines morphological complexity with underrepresentation in pre-training data. This has direct cost implications: processing the same EU regulation in Ukrainian costs $3\times$ more than in English, purely due to tokenizer design.

5.2 Few-Shot Degradation Is Task-Dependent, Not Language-Intrinsic

Experiment 2 reveals that the few-shot effect is neither uniformly beneficial nor uniformly harmful on Ukrainian. On SIB-200 news classification, five of seven models *improve* with few-shot examples – the reverse of legal text, where five of seven degraded. The most dramatic reversals are Nemotron Super 3 (-12.8 pp legal, $+8.3$ pp news) and Qwen3 235B (-26.0 pp legal, $+4.9$ pp news).

Crucially, two models – Llama 4 Maverick (-7.8 pp) and Mistral Large 3 (-4.4 pp) – degrade on *both* tasks, revealing a model-level vulnerability to few-shot interference that persists across domains. This three-way decomposition (model effect $\times$ task effect $\times$ language) replaces the simpler “Ukrainian few-shot degradation” narrative: **few-shot demonstrations help when the task has balanced, diverse categories and the model can leverage them for format specification; they hurt when the task has imbalanced, formulaic outputs that cause anchoring, or when the model’s attention mechanism is vulnerable to surface-pattern matching regardless of task.**

For practitioners, few-shot validation must be performed *per model and per task*, not per language. A model that benefits from few-shot on topic classification may catastrophically degrade on outcome classification in the same language.

5.3 Morphological Complexity vs. Undertrained Language

5.4 Fertility Predicts Linguistic Competence

5.5 Practical Recommendations

1. **Default to zero-shot** for morphologically rich languages. Validate few-shot per model and task before deploying.

2. **Start with tokenizer analysis.** Fertility rankings are domain-invariant; measure once, apply everywhere.
3. **Ignore parameter counts** for non-English model selection. Use language-specific benchmarks.
4. **Budget for the tokenizer tax.** A model with $1.6\times$ higher fertility is $1.6\times$ more expensive per document.

6 Limitations

SIB-200 test set size. The test split contains only 204 examples with 7 classes, resulting in some classes having as few as 17 instances (geography). Confidence intervals on minority classes are wide.

Slavic language coverage. We test four Slavic languages. Including non-Slavic morphologically rich languages (e.g., Finnish, Hungarian, Turkish) would strengthen the morphological complexity hypothesis.

TMX segment quality. EU Act translations may contain formulaic boilerplate that inflates cross-lingual similarity and deflates fertility differences.

ULP dataset size. With 347 questions and only 7 data points in the fertility–competence correlation, statistical power is limited.

Same models as Paper 1. We use the same seven models for comparability, but the landscape evolves rapidly. Results may not generalize to models released after May 2026.

7 Conclusion

Acknowledgments

This work was conducted as part of the LEX AI platform development at legal.org.ua. Compute costs for all experiments were covered by an AWS Activate grant. We thank the creators of SIB-200, the ULP benchmark, and the EU Acts in Ukrainian corpus for making their datasets publicly available.

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A EU Acts Fertility: Full Results